

Profit Dilution Resistance and the Cross Section of Stock Returns

I. M. Harking

October 20, 2025

Abstract

This paper studies the asset pricing implications of Profit Dilution Resistance (PDR), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023a\)](#). A value-weighted long/short trading strategy based on PDR achieves an annualized gross (net) Sharpe ratio of 0.52 (0.45), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 24 (23) bps/month with a t-statistic of 3.12 (3.01), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Growth in book equity, Share issuance (5 year), Share issuance (1 year), Change in equity to assets, Momentum and LT Reversal, Long-run reversal) is 14 bps/month with a t-statistic of 1.99.

1 Introduction

Financial markets continuously process vast amounts of information, yet the speed and completeness of this incorporation remains a central question in asset pricing. While the efficient market hypothesis suggests that prices should rapidly reflect all available information, mounting evidence indicates that information diffuses gradually through markets, creating predictable patterns in returns. This slow diffusion manifests particularly in firms' responses to changes in their capital structure and profitability dynamics, where complex accounting relationships may obscure the full implications for future cash flows.

We examine how markets incorporate information about firms' ability to maintain profitability in the face of equity dilution—a phenomenon we term Profit Dilution Resistance (PDR). When firms issue new equity or experience changes in their capital base, their ability to sustain or grow profits per share becomes crucial for valuation, yet this information appears to be only gradually recognized by market participants. This study investigates whether PDR predicts cross-sectional stock returns and whether this predictability concentrates in market segments where information frictions are most severe.

The gradual diffusion of information provides a natural framework for understanding why PDR might predict returns. Following the seminal work of [Hong et al. \(2000\)](#) on the slow diffusion of industry information and [DellaVigna and Pollet \(2009\)](#) on investor inattention, we posit that investors systematically underreact to signals about firms' profit dilution resistance. When firms demonstrate an ability to maintain or grow profitability despite equity base changes, this positive information incorporates into prices only gradually, leading to predictable return continuation. This mechanism aligns with [Hirshleifer et al. \(2009\)](#), who document that investors have limited attention and fail to fully process information with complex implications for firm value.

The underreaction to PDR information likely stems from its computational complexity and its embedding within detailed financial statements. [Cohen et al. \(2003\)](#) show that investors often overlook information that requires combining multiple accounting variables, while [Richardson et al. \(2003\)](#) demonstrate systematic underreaction to information in accruals and related accounting measures. PDR requires investors to jointly evaluate changes in equity base alongside profit dynamics—a multidimensional assessment that exceeds the processing capacity of many market participants, particularly retail investors and those with limited attention. This complexity-induced neglect creates temporary mispricing that resolves as information gradually diffuses through the market.

We predict that PDR’s return predictability will concentrate among stocks with greater information frictions. [Hong et al. \(2007\)](#) and [Hou \(2007\)](#) establish that information diffuses more slowly for small stocks, those with low analyst coverage, and those with limited institutional ownership. Similarly, [Peng and Xiong \(2007\)](#) show that investors with limited attention focus on market-level information at the expense of firm-specific signals, particularly for smaller, less visible firms. These findings suggest that PDR signals will generate stronger return continuation among neglected stocks where attention constraints bind most tightly and where sophisticated arbitrageurs face the greatest limits to arbitrage.

Our empirical analysis reveals that PDR strongly predicts cross-sectional stock returns. A value-weighted long/short trading strategy based on PDR achieves an annualized gross Sharpe ratio of 0.52, placing it in the top 10% of documented anomalies. The strategy generates a monthly average excess return of 31 basis points with a t-statistic of 4.05, and this performance remains robust after controlling for common risk factors. The monthly alpha relative to the Fama-French six-factor model equals 24 basis points with a t-statistic of 3.12, indicating that PDR captures a distinct source of mispricing beyond established factors.

The economic magnitude of PDR’s predictive power remains substantial across various portfolio construction methodologies and after accounting for transaction costs. Net of trading costs, the value-weighted quintile strategy delivers an annualized Sharpe ratio of 0.45 and a monthly alpha of 23 basis points relative to the six-factor model (t -statistic = 3.01). The strategy’s performance persists across different sorting procedures: using equal-weighting yields a net alpha of 20 basis points per month, while sorts based on name breaks and market capitalization breaks produce net alphas of 26 and 20 basis points, respectively. These results demonstrate that PDR’s return predictability is not an artifact of specific portfolio construction choices.

Consistent with the slow diffusion hypothesis, PDR’s predictive power concentrates among stocks where information frictions are most severe. Among the smallest quintile of stocks, the PDR strategy generates an average monthly return of 50 basis points with a t -statistic of 5.01 and a six-factor alpha of 51 basis points. Even among the largest quintile of stocks—where information should diffuse most rapidly—the strategy achieves a statistically significant return of 23 basis points per month (t -statistic = 2.57) and a six-factor alpha of 20 basis points. This size-conditional evidence supports our hypothesis that PDR captures gradual information incorporation, with the strongest effects where attention constraints and information processing limitations are most binding.

This paper contributes to the literature on information diffusion and cross-sectional return predictability in several important ways. While [Pontiff and Woodgate \(2008\)](#) examine share issuance and its return implications, and [Daniel and Titman \(2006\)](#) study the market’s reaction to changes in book equity, our focus on profit dilution resistance provides a more nuanced measure that captures firms’ fundamental ability to create value despite equity base changes. Unlike simple issuance-based measures studied in [Loughran and Ritter \(1995\)](#) and [Spiess and Affleck-Graves \(1995\)](#), PDR

incorporates the crucial dimension of concurrent profitability dynamics, offering a more complete picture of how dilution affects shareholder value. Our results show that controlling for related anomalies including five-year share issuance, one-year share issuance, and growth in book equity, PDR retains significant predictive power with an alpha of 14 basis points per month (t-statistic = 1.99).

Methodologically, we advance the literature by providing comprehensive evidence on the robustness of PDR across multiple dimensions. Following the protocol of [Novy-Marx and Velikov \(2023b\)](#), we demonstrate that PDR’s predictive power survives extensive robustness tests including alternative portfolio construction methods, double sorts on size, and controls for 210 other documented anomalies. Our analysis of transaction costs using the [Chen and Velikov \(2022b\)](#) composite bid-ask spread measure shows that PDR remains profitable after implementation costs, with net Sharpe ratios ranking in the top 2% of all anomalies. This comprehensive approach addresses concerns raised by [Harvey et al. \(2016\)](#) about data mining and [Hou et al. \(2020\)](#) about replication failures in the anomalies literature.

Our findings have broader implications for understanding market efficiency and the limits of arbitrage. The persistence of PDR’s predictive power, even among large-cap stocks with substantial analyst coverage, suggests that information complexity creates significant barriers to efficient price discovery. This evidence supports theoretical models of limited attention such as [Peng and Xiong \(2006\)](#) and [Kacperczyk et al. \(2016\)](#), where investors must allocate scarce cognitive resources across multiple information sources. By identifying PDR as a robust predictor linked to gradual information diffusion, we provide new evidence on how accounting complexity and multidimensional firm characteristics can generate persistent mispricing in modern financial markets.

2 Data

We obtain financial statement data from the COMPUSTAT North America database, which provides comprehensive accounting information for publicly traded U.S. companies. Our sample includes all firms with available data in COMPUSTAT, excluding financial firms (SIC codes 6000-6999) and utilities (SIC codes 4900-4999) due to their distinct regulatory environments and capital structures. The sample spans multiple decades of annual observations, allowing us to examine the signal's performance across various market conditions and business cycles. We construct our signal using two key accounting variables from firms' annual financial statements. Common stock (CSTK) represents the par or stated value of common stock outstanding as reported on the balance sheet, reflecting the nominal value assigned to issued common shares. Gross profit (GP) measures the difference between total revenues and cost of goods sold, representing the fundamental profitability of a firm's core operations before considering operating expenses, interest, and taxes. Both variables are measured in millions of dollars and obtained from firms' fiscal year-end financial statements. We construct the Profit Dilution Resistance signal by computing the year-over-year change in common stock scaled by lagged gross profit: $(\text{CSTK}(t) - \text{CSTK}(t-1)) / \text{GP}(t-1)$. This ratio captures the relationship between changes in equity capital structure and the firm's ability to generate profits from its core operations. A higher value indicates that a firm is issuing common stock at a rate that exceeds its profit-generating capacity, potentially signaling dilution of existing shareholders' claims on future profits. We calculate this measure annually using fiscal year-end values and require non-missing values for both common stock in consecutive years and lagged gross profit. To mitigate the influence of outliers, we winsorize the signal at the 1st and 99th percentiles and exclude observations where lagged gross profit is negative or zero, as these cases would produce economically meaningless scaling.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the PDR signal. Panel A plots the time-series of the mean, median, and interquartile range for PDR. On average, the cross-sectional mean (median) PDR is -0.03 (-0.00) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input PDR data. The signal’s interquartile range spans -0.02 to 0.00. Panel B of Figure 1 plots the time-series of the coverage of the PDR signal for the CRSP universe. On average, the PDR signal is available for 6.28% of CRSP names, which on average make up 7.67% of total market capitalization.

4 Does PDR predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on PDR using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high PDR portfolio and sells the low PDR portfolio. The rest of Panel A reports the portfolios’ monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short PDR strategy earns an average return of 0.31% per month with a t-statistic of 4.05. The annualized Sharpe ratio of the strategy is 0.52. The alphas range from 0.24% to 0.32% per month and have t-statistics exceeding 3.12 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios’ loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy’s most significant loading is 0.28,

with a t-statistic of 5.29 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 572 stocks and an average market capitalization of at least \$1,523 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor’s achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using cap breakpoints and value-weighted portfolios, and equals 26 bps/month with a t-statistics of 3.45. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-five exceed two, and for twenty-one exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022a\)](#). The net average returns

reported in the first column range between 22-29bps/month. The lowest return, (22 bps/month), is achieved from the quintile sort using cap breakpoints and value-weighted portfolios, and has an associated t-statistic of 2.97. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the PDR trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-five cases, and significantly expands the achievable frontier in twenty-five cases.

Table 3 provides direct tests for the role size plays in the PDR strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and PDR, as well as average returns and alphas for long/short trading PDR strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the PDR strategy achieves an average return of 23 bps/month with a t-statistic of 2.57. Among these large cap stocks, the alphas for the PDR strategy relative to the five most common factor models range from 20 to 23 bps/month with t-statistics between 2.11 and 2.50.

5 How does PDR perform relative to the zoo?

Figure 2 puts the performance of PDR in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the PDR strategy falls in the distribution. The PDR strategy’s gross (net) Sharpe ratio of 0.52 (0.45) is greater than 90% (98%) of anomaly Sharpe ratios,

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the PDR strategy (red line).² Ignoring trading costs, a \$1 invested in the PDR strategy would have yielded \$7.23 which ranks the PDR strategy in the top 2% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the PDR strategy would have yielded \$5.25 which ranks the PDR strategy in the top 1% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the PDR relative to those. Panel A shows that the PDR strategy gross alphas fall between the 61 and 73 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The PDR strategy has a positive net generalized alpha for five out of the five factor models. In these cases PDR ranks between the 80 and 88 percentiles in terms of how much it could have expanded the achievable investment frontier.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in Detzel et al. (2022), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

6 Does PDR add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of PDR with 210 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price PDR or at least to weaken the power PDR has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of PDR conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{PDR} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{PDR}PDR_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{PDR,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on PDR. Stocks are finally grouped into five PDR portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

PDR trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on PDR and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the PDR signal in these Fama-MacBeth regressions exceed 0.82, with the minimum t-statistic occurring when controlling for Momentum and LT Reversal. Controlling for all six closely related anomalies, the t-statistic on PDR is 0.00.

Similarly, Table 5 reports results from spanning tests that regress returns to the PDR strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the PDR strategy earns alphas that range from 16-25bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.16, which is achieved when controlling for Momentum and LT Reversal. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the PDR trading strategy achieves an alpha of 14bps/month with a t-statistic of 1.99.

7 Does PDR add relative to the whole zoo?

Finally, we can ask how much adding PDR to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022a\)](#). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the PDR signal.⁴ We consider

⁴We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capital-

one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 154-anomaly combination strategy grows to \$2560.30, while \$1 investment in the combination strategy that includes PDR grows to \$3089.32.

8 Conclusion

This study provides compelling evidence that Profit Dilution Resistance (PDR) represents a economically meaningful and statistically significant predictor of cross-sectional equity returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that PDR-based trading strategies generate substantial risk-adjusted returns that persist even after accounting for well-established factors and related anomalies.

The key findings reveal that a value-weighted long/short strategy based on PDR delivers an annualized Sharpe ratio of 0.52 gross (0.45 net of transaction costs), indicating robust performance that survives practical implementation considerations. More importantly, the strategy generates significant alpha relative to the Fama-French five-factor model augmented with momentum, yielding monthly abnormal returns of 24 basis points gross (23 basis points net) with t-statistics exceeding 3.0. Even when controlling for six closely related strategies from the factor zoo—including various measures of share issuance, equity growth, and reversal effects—the PDR signal maintains a positive alpha of 14 basis points per month with a t-statistic of 1.99, suggesting that it captures unique information about future returns not fully

ization on CRSP in the period for which PDR is available.

subsumed by existing factors.

From a practical perspective, these results suggest that PDR can serve as a valuable signal for portfolio construction and risk management. The strategy’s ability to generate positive net returns after transaction costs indicates its viability for real-world implementation, while its persistence after controlling for related factors suggests it provides incremental value to existing factor models. The economic magnitude of the returns, combined with their statistical significance, positions PDR as a meaningful addition to the toolkit of quantitative investors and asset managers.

However, several limitations warrant consideration. First, while our analysis follows best practices for anomaly testing, the proliferation of documented anomalies raises concerns about data mining and multiple testing. Future research should examine the out-of-sample performance of PDR, particularly in international markets where data availability permits. Second, the economic mechanism underlying PDR’s predictive power deserves further investigation. Understanding whether the signal reflects risk compensation, behavioral biases, or market frictions would enhance our theoretical understanding and inform practical applications.

Future research directions include examining the time-varying nature of PDR’s effectiveness, particularly during different market regimes and economic cycles. Additionally, investigating potential interactions between PDR and firm characteristics such as size, profitability, and financial constraints could reveal important heterogeneity in the signal’s effectiveness. Finally, exploring the relationship between PDR and corporate actions, such as MA activity or capital structure decisions, may provide insights into the fundamental drivers of the anomaly and its persistence in increasingly efficient markets.

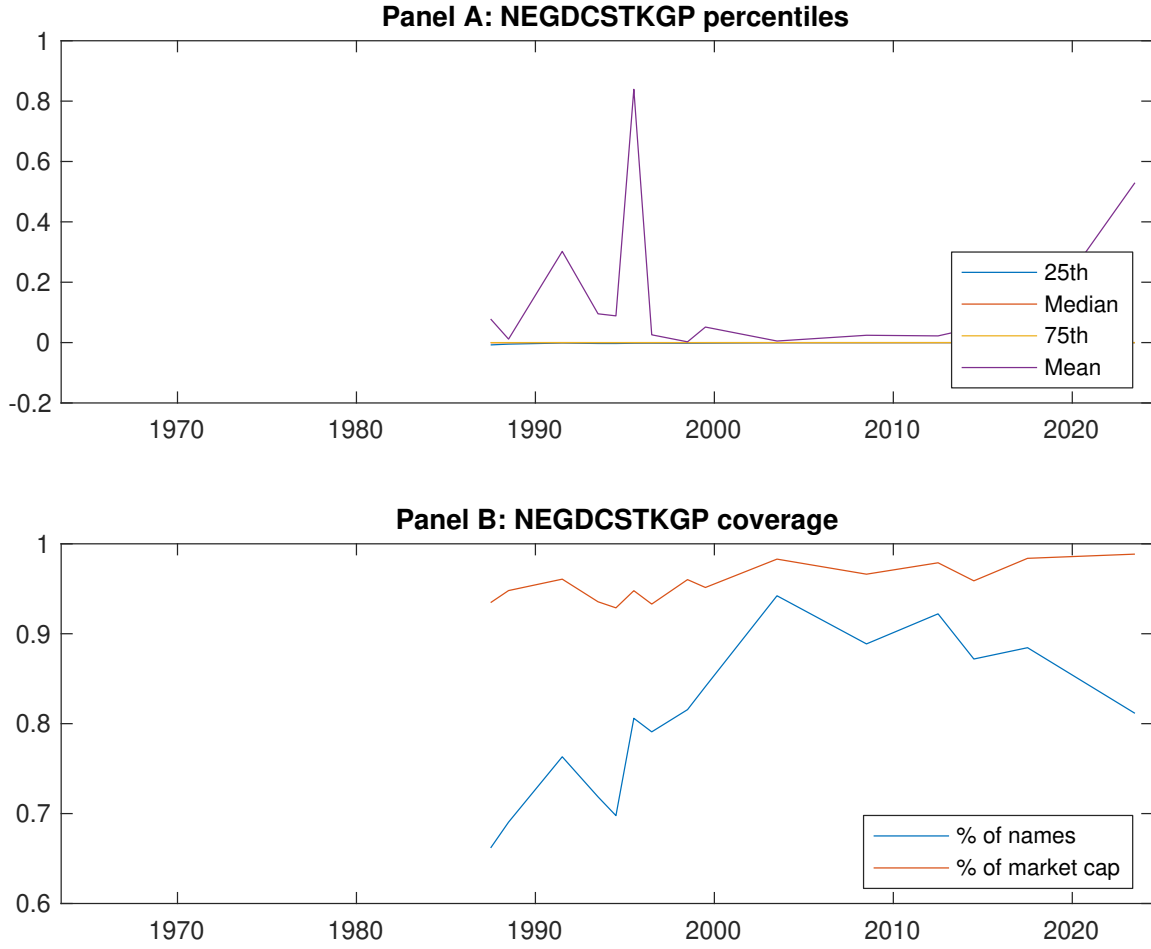


Figure 1: Times series of PDR percentiles and coverage. This figure plots descriptive statistics for PDR. Panel A shows cross-sectional percentiles of PDR over the sample. Panel B plots the monthly coverage of PDR relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on PDR. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on PDR-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.44 [2.63]	0.55 [3.00]	0.61 [3.43]	0.64 [3.99]	0.74 [4.62]	0.31 [4.05]
α_{CAPM}	-0.12 [-2.31]	-0.08 [-1.86]	-0.00 [-0.07]	0.10 [1.97]	0.19 [4.35]	0.32 [4.13]
α_{FF3}	-0.15 [-2.76]	-0.06 [-1.44]	0.02 [0.37]	0.06 [1.23]	0.16 [3.61]	0.30 [3.92]
α_{FF4}	-0.12 [-2.26]	-0.04 [-0.81]	0.04 [0.89]	0.03 [0.58]	0.14 [3.30]	0.27 [3.41]
α_{FF5}	-0.17 [-3.26]	0.00 [0.11]	0.02 [0.49]	-0.02 [-0.54]	0.09 [2.20]	0.27 [3.47]
α_{FF6}	-0.15 [-2.83]	0.02 [0.50]	0.04 [0.92]	-0.04 [-0.93]	0.09 [2.11]	0.24 [3.12]
Panel B: Fama and French (2018) 6-factor model loadings for PDR-sorted portfolios						
β_{MKT}	0.97 [75.11]	1.04 [96.77]	1.02 [91.08]	1.00 [91.09]	0.98 [93.93]	0.01 [0.59]
β_{SMB}	-0.03 [-1.82]	0.00 [0.22]	0.03 [1.69]	-0.06 [-3.98]	0.01 [0.61]	0.04 [1.59]
β_{HML}	0.10 [3.88]	-0.02 [-0.78]	-0.06 [-2.96]	0.06 [3.04]	0.01 [0.52]	-0.09 [-2.38]
β_{RMW}	0.13 [5.10]	-0.13 [-5.96]	0.01 [0.27]	0.13 [5.89]	0.06 [3.15]	-0.06 [-1.75]
β_{CMA}	-0.07 [-1.98]	-0.12 [-3.86]	-0.02 [-0.63]	0.17 [5.55]	0.21 [7.06]	0.28 [5.29]
β_{UMD}	-0.03 [-2.39]	-0.02 [-2.32]	-0.03 [-2.61]	0.03 [2.34]	0.00 [0.33]	0.03 [1.83]
Panel C: Average number of firms (n) and market capitalization (me)						
n	684	652	572	671	803	
me (\$10 ⁶)	1688	1523	2173	2213	2520	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the PDR strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.31 [4.05]	0.32 [4.13]	0.30 [3.92]	0.27 [3.41]	0.27 [3.47]	0.24 [3.12]
Quintile	NYSE	EW	0.44 [7.75]	0.46 [7.96]	0.43 [7.80]	0.38 [6.80]	0.41 [7.48]	0.37 [6.72]
Quintile	Name	VW	0.33 [4.18]	0.34 [4.31]	0.33 [4.14]	0.30 [3.68]	0.30 [3.73]	0.28 [3.42]
Quintile	Cap	VW	0.26 [3.45]	0.26 [3.48]	0.26 [3.40]	0.21 [2.78]	0.24 [3.24]	0.21 [2.78]
Decile	NYSE	VW	0.33 [3.75]	0.30 [3.46]	0.29 [3.26]	0.24 [2.73]	0.28 [3.15]	0.24 [2.76]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.27 [3.56]	0.28 [3.67]	0.26 [3.46]	0.24 [3.18]	0.24 [3.20]	0.23 [3.01]
Quintile	NYSE	EW	0.25 [3.94]	0.28 [4.36]	0.25 [3.99]	0.22 [3.59]	0.21 [3.55]	0.20 [3.27]
Quintile	Name	VW	0.29 [3.70]	0.30 [3.90]	0.29 [3.71]	0.27 [3.46]	0.27 [3.47]	0.26 [3.31]
Quintile	Cap	VW	0.22 [2.97]	0.22 [3.03]	0.22 [2.93]	0.19 [2.60]	0.21 [2.94]	0.20 [2.69]
Decile	NYSE	VW	0.28 [3.26]	0.26 [2.94]	0.24 [2.76]	0.22 [2.49]	0.23 [2.72]	0.22 [2.52]

Table 3: Conditional sort on size and PDR

This table presents results for conditional double sorts on size and PDR. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on PDR. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high PDR and short stocks with low PDR. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 1963:06 to 2024:06.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	PDR Quintiles					PDR Strategies						
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
	(1)	0.47 [1.95]	0.77 [3.10]	0.98 [3.86]	0.92 [3.85]	0.95 [3.73]	0.50 [5.01]	0.50 [4.96]	0.54 [5.48]	0.42 [4.34]	0.61 [6.32]	0.51 [5.37]
	(2)	0.63 [2.84]	0.77 [3.31]	0.89 [3.80]	0.94 [4.24]	0.84 [3.79]	0.21 [2.48]	0.23 [2.66]	0.21 [2.53]	0.17 [1.98]	0.26 [3.06]	0.22 [2.60]
	(3)	0.55 [2.79]	0.67 [3.12]	0.81 [3.74]	0.81 [4.01]	0.90 [4.53]	0.35 [4.59]	0.34 [4.45]	0.34 [4.39]	0.30 [3.76]	0.35 [4.48]	0.32 [3.96]
	(4)	0.47 [2.52]	0.67 [3.34]	0.81 [4.06]	0.75 [4.00]	0.80 [4.37]	0.33 [4.51]	0.34 [4.64]	0.31 [4.20]	0.30 [4.03]	0.19 [2.68]	0.20 [2.75]
	(5)	0.47 [2.89]	0.49 [2.70]	0.55 [3.15]	0.50 [3.07]	0.70 [4.42]	0.23 [2.57]	0.23 [2.50]	0.23 [2.47]	0.20 [2.11]	0.22 [2.42]	0.20 [2.16]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	PDR Quintiles					PDR Quintiles						
		Average n					Average market capitalization (\$10 ⁶)					
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)	
	(1)	375	376	374	373	366	31	33	38	28	29	
	(2)	108	106	107	107	105	57	55	57	56	54	
	(3)	78	77	77	77	77	98	95	98	99	98	
	(4)	65	65	65	65	65	205	206	216	210	214	
(5)	60	60	60	60	60	1447	1524	1677	1622	1876		

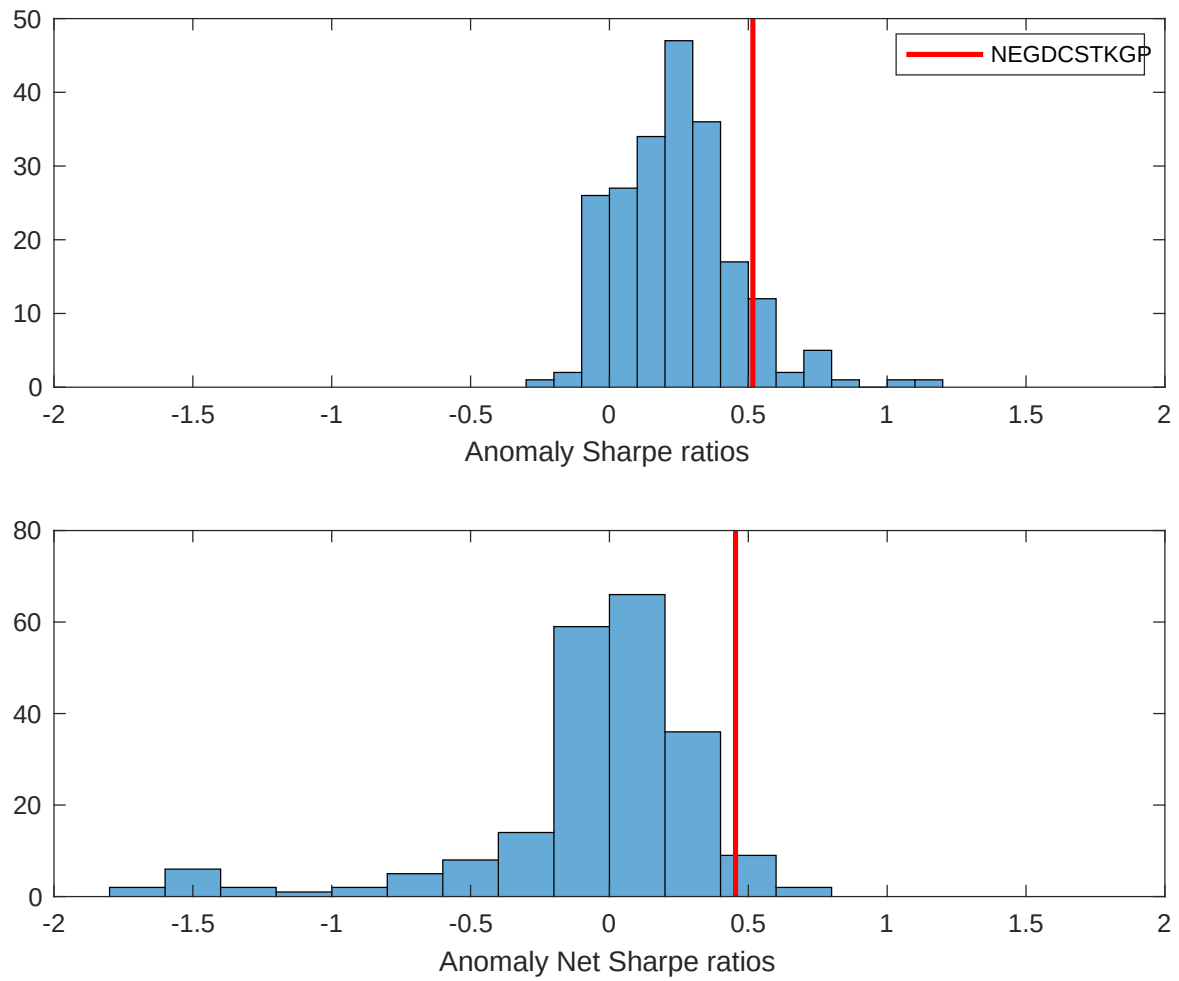


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the PDR with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

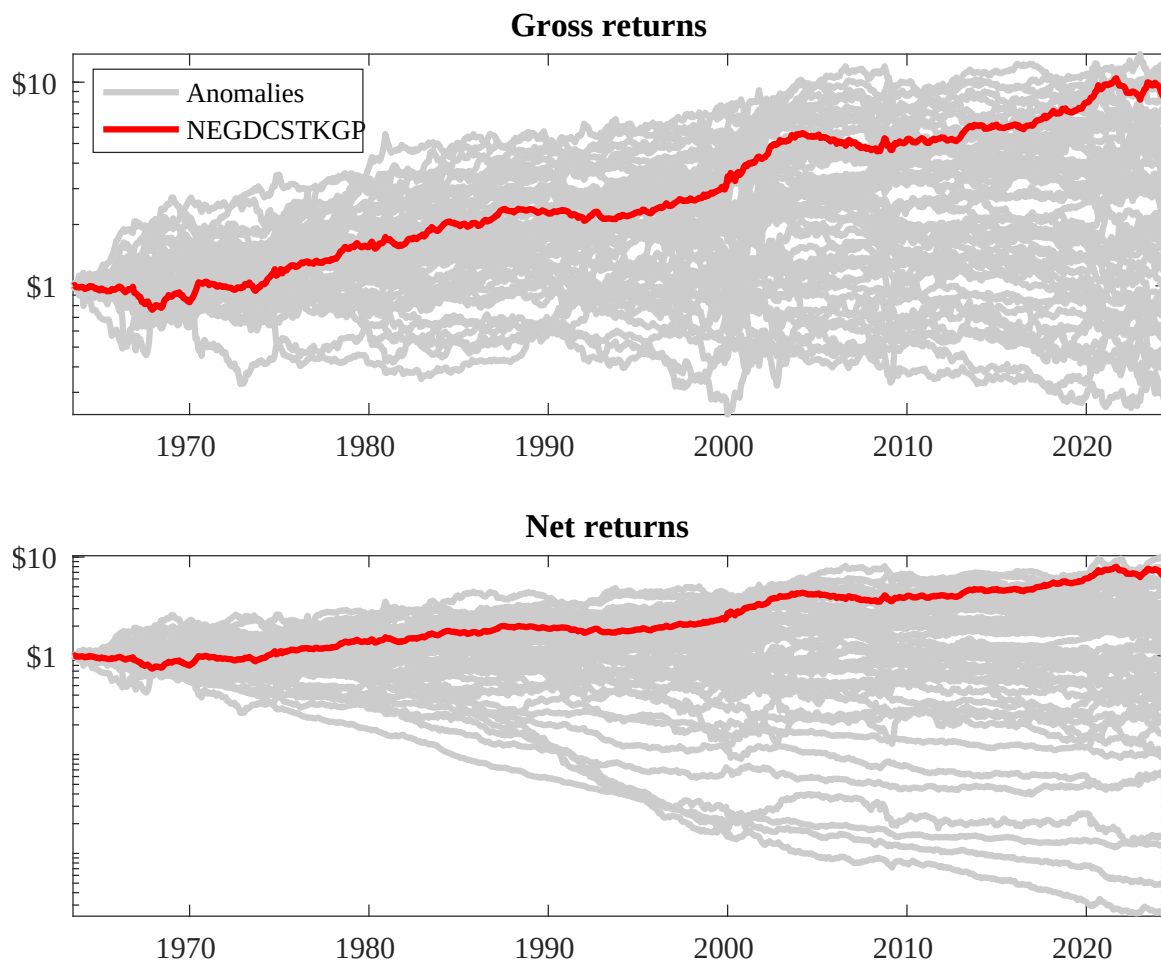
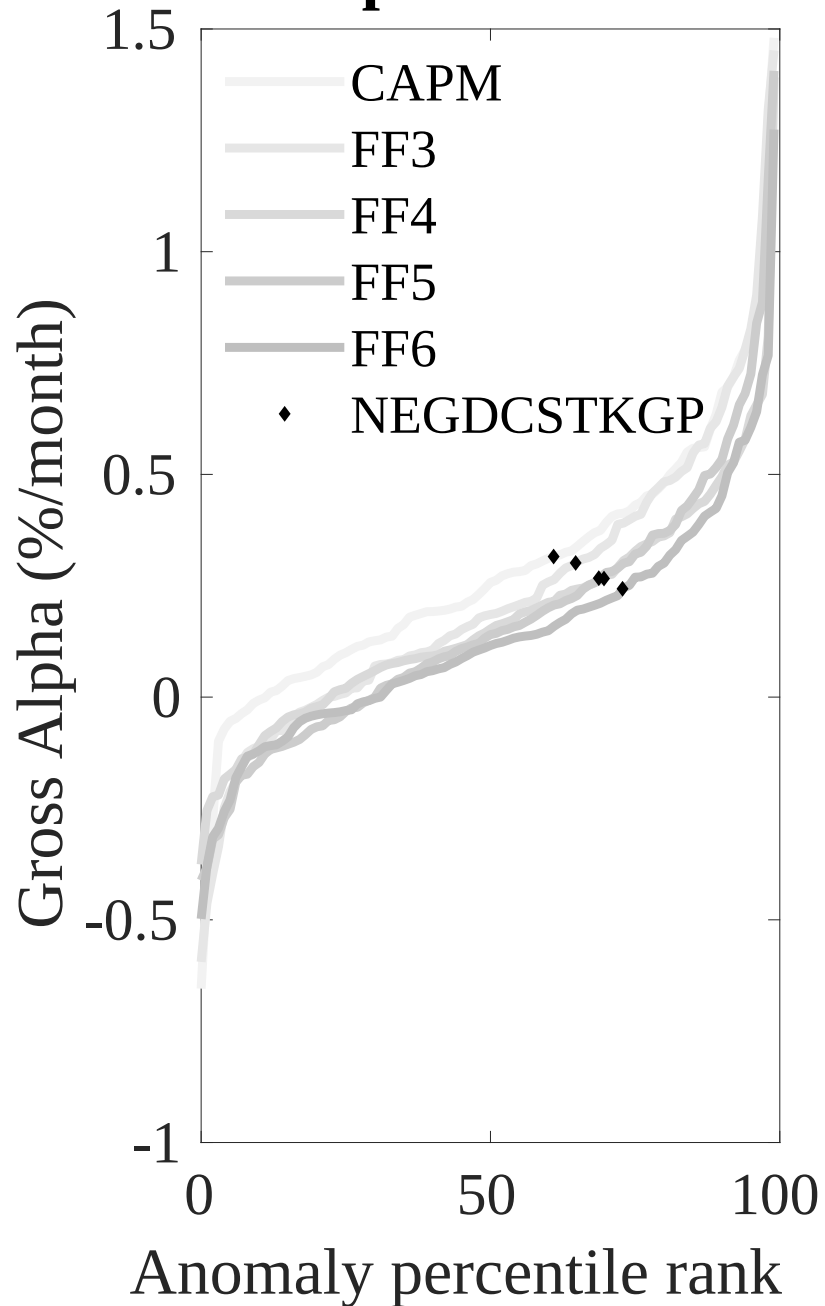


Figure 3: Dollar invested.

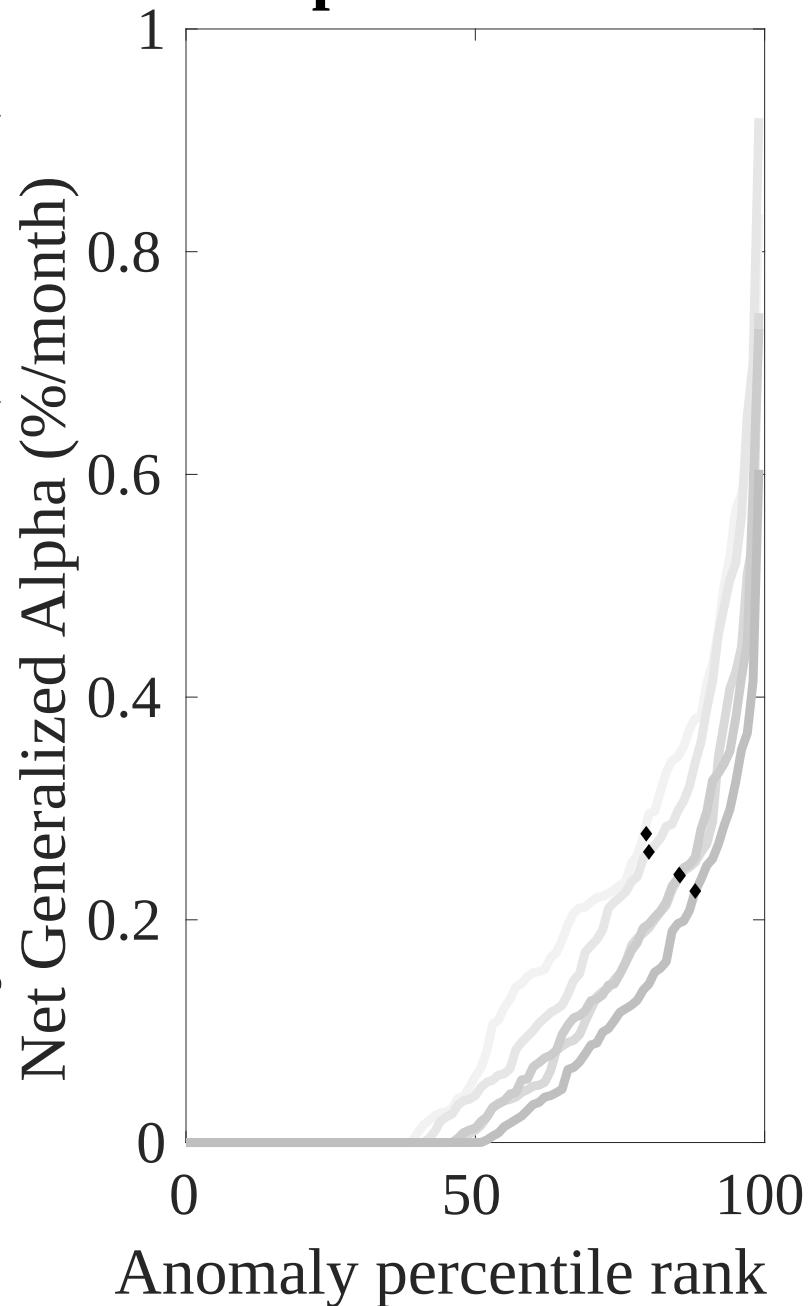
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the PDR trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Net Alpha Percentiles

Novy-Marx and Velikov (RFS, 2016)



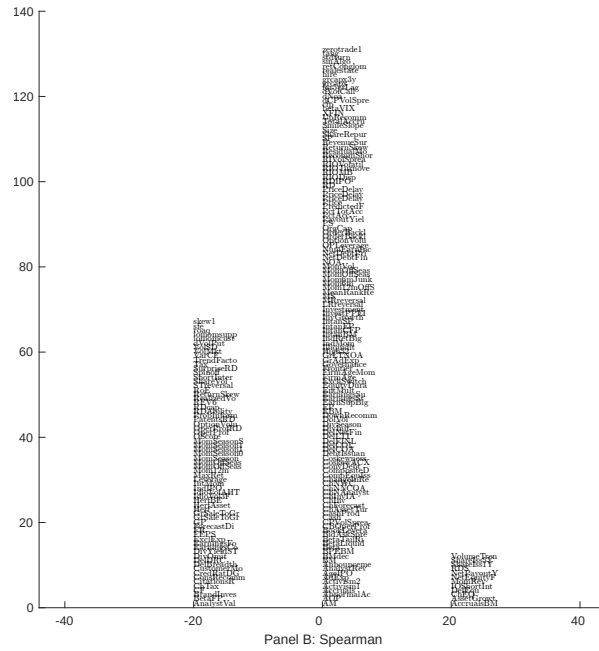
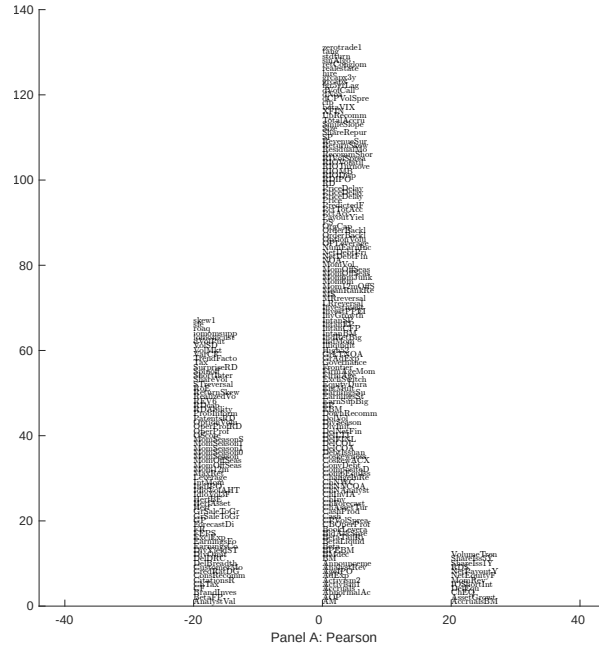


Figure 5: Distribution of correlations.

This figure plots a name histogram of correlations of 210 filtered anomaly signals with PDR. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

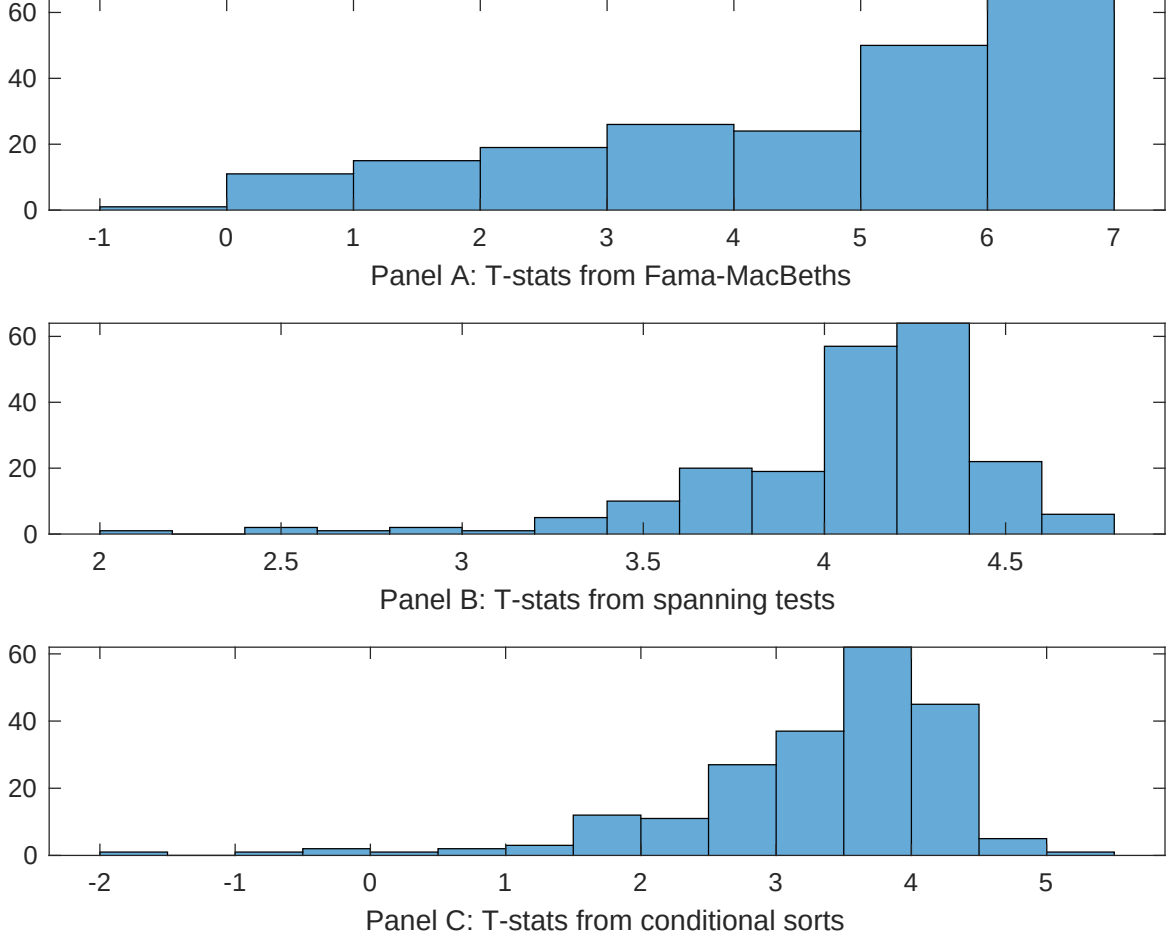


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of PDR conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{PDR} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{PDR}PDR_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{PDR,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on PDR. Stocks are finally grouped into five PDR portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted PDR trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on PDR. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{PDR}PDR_{i,t} + \sum_{k=1}^s \beta_{X_k} X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Growth in book equity, Share issuance (5 year), Share issuance (1 year), Change in equity to assets, Momentum and LT Reversal, Long-run reversal. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.17 [6.93]	0.13 [6.46]	0.13 [6.09]	0.12 [5.77]	0.40 [1.19]	0.12 [5.83]	0.10 [2.50]
PDR	0.14 [4.69]	0.11 [3.80]	0.12 [4.45]	0.15 [5.52]	0.10 [0.82]	0.14 [5.57]	0.26 [0.00]
Anomaly 1	0.38 [3.19]						0.30 [1.25]
Anomaly 2		0.26 [4.76]					0.28 [2.83]
Anomaly 3			0.13 [5.79]				0.11 [2.21]
Anomaly 4				0.12 [3.23]			-0.11 [-1.46]
Anomaly 5					0.11 [4.24]		0.97 [3.01]
Anomaly 6						0.22 [2.49]	-0.26 [-0.20]
# months	720	715	715	720	648	715	600
$\bar{R}^2(\%)$	0	0	0	0	2	1	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the PDR trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{PDR} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Growth in book equity, Share issuance (5 year), Share issuance (1 year), Change in equity to assets, Momentum and LT Reversal, Long-run reversal. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.23 [3.06]	0.16 [2.16]	0.23 [3.03]	0.24 [3.15]	0.22 [2.88]	0.25 [3.24]	0.14 [1.99]
Anomaly 1	29.44 [7.27]						29.60 [5.12]
Anomaly 2		32.34 [8.15]					20.89 [4.57]
Anomaly 3			28.17 [7.44]				11.92 [2.70]
Anomaly 4				11.17 [2.93]			-13.71 [-2.65]
Anomaly 5					4.80 [4.61]		3.95 [3.71]
Anomaly 6						5.07 [2.24]	-0.34 [-0.14]
mkt	2.27 [1.27]	2.73 [1.54]	1.99 [1.12]	1.10 [0.60]	1.45 [0.80]	1.04 [0.56]	3.82 [2.21]
smb	4.58 [1.78]	7.70 [2.99]	6.63 [2.57]	5.42 [2.04]	3.62 [1.36]	3.28 [1.17]	5.38 [2.07]
hml	-10.75 [-3.09]	-4.68 [-1.37]	-7.96 [-2.31]	-8.60 [-2.40]	-8.36 [-2.37]	-8.65 [-2.37]	-8.93 [-2.63]
rmw	-5.06 [-1.45]	-18.09 [-4.84]	-21.74 [-5.38]	-5.43 [-1.51]	-4.01 [-1.12]	-4.33 [-1.17]	-18.44 [-4.54]
cma	-1.47 [-0.23]	11.43 [2.13]	9.58 [1.72]	15.80 [2.44]	24.69 [4.76]	23.10 [4.22]	-6.78 [-1.04]
umd	3.08 [1.74]	1.47 [0.83]	2.30 [1.30]	3.88 [2.13]	-1.08 [-0.52]	3.90 [2.14]	-3.07 [-1.52]
# months	732	728	728	732	728	728	728
$\bar{R}^2(\%)$	12	13	12	7	8	6	20

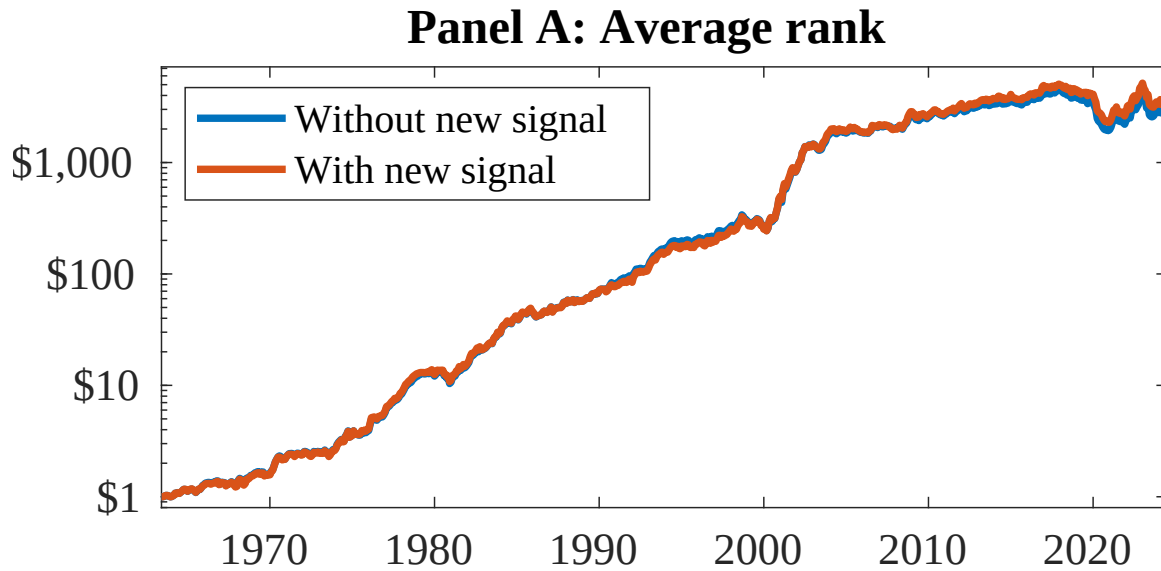


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022a\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as PDR. Panel A shows results using "Average rank" as the combination method. See [Section 7](#) for details on the combination methods.

References

- Carhart, M. M. (1997). On persistence in mutual fund performance. *Journal of Finance*, 52:57–82.
- Chen, A. and Velikov, M. (2022a). Zeroing in on the expected returns of anomalies. *Journal of Financial and Quantitative Analysis*, Forthcoming.
- Chen, A. Y. and Velikov, M. (2022b). High-frequency measures of market liquidity. *Working Paper*.
- Chen, A. Y. and Zimmermann, T. (2022). Open source cross-sectional asset pricing. *Critical Finance Review*, 27(2):207–264.
- Cohen, D. A., Dey, A., and Lys, T. Z. (2003). The value relevance of components of other comprehensive income when net income is disaggregated. *Journal of Accounting Research*, 41(5):775–804.
- Daniel, K. and Titman, S. (2006). Market reactions to tangible and intangible information. *Journal of Finance*, 61(4):1605–1643.
- DellaVigna, S. and Pollet, J. M. (2009). Investor inattention and friday earnings announcements. *Journal of Finance*, 64(2):709–749.
- Detzel, A., Novy-Marx, R., and Velikov, M. (2022). Model comparison with transaction costs. *Journal of Finance*, Forthcoming.
- Fama, E. F. and French, K. R. (1993). Common risk factors in the returns on stocks and bonds. *Journal of Financial Economics*, 33(1):3–56.
- Fama, E. F. and French, K. R. (2015). A five-factor asset pricing model. *Journal of Financial Economics*, 116(1):1–22.

- Fama, E. F. and French, K. R. (2018). Choosing factors. *Journal of Financial Economics*, 128(2):234–252.
- Harvey, C. R., Liu, Y., and Zhu, H. (2016). ...and the cross-section of expected returns. *Review of Financial Studies*, 29(1):5–68.
- Hirshleifer, D., Lim, S. S., and Teoh, S. H. (2009). Driven to distraction: Extraneous events and underreaction to earnings news. *Journal of Finance*, 64(5):2289–2325.
- Hong, H., Lim, T., and Stein, J. C. (2000). Bad news travels slowly: Size, analyst coverage, and the profitability of momentum strategies. *Journal of Finance*, 55(1):265–295.
- Hong, H., Torous, W., and Valkanov, R. (2007). Industry information diffusion and the lead-lag effect in stock returns. *Review of Financial Studies*, 20(4):1113–1138.
- Hou, K. (2007). Industry information diffusion and the lead-lag effect in stock returns. *Review of Financial Studies*, 20(4):1113–1138.
- Hou, K., Xue, C., and Zhang, L. (2020). Replicating anomalies. *Review of Financial Studies*, 33(5):2019–2133.
- Kacperczyk, M., Van Nieuwerburgh, S., and Veldkamp, L. (2016). A rational theory of mutual funds’ attention allocation. *Journal of Finance*, 71(2):661–698.
- Loughran, T. and Ritter, J. R. (1995). The new issues puzzle. *Journal of Finance*, 50(1):23–51.
- Novy-Marx, R. and Velikov, M. (2016). A taxonomy of anomalies and their trading costs. *Review of Financial Studies*, 29(1):104–147.
- Novy-Marx, R. and Velikov, M. (2023a). Assaying anomalies. *Working paper*.
- Novy-Marx, R. and Velikov, M. (2023b). Assaying anomalies. *Working Paper*.

- Peng, L. and Xiong, W. (2006). Investor attention and time-varying comovements. *Journal of Financial Economics*, 80(3):563–602.
- Peng, L. and Xiong, W. (2007). Investor attention, overconfidence and category learning. *Journal of Financial Economics*, 80(3):563–602.
- Pontiff, J. and Woodgate, A. (2008). Share issuance and cross-sectional returns. *Journal of Finance*, 63(2):921–945.
- Richardson, S. A., Sloan, R. G., Soliman, M. T., and Tuna, I. (2003). Accrual reliability, earnings persistence and stock prices. *Journal of Accounting and Economics*, 39(3):437–485.
- Spiess, D. K. and Affleck-Graves, J. (1995). Underperformance in long-run stock returns following seasoned equity offerings. *Journal of Financial Economics*, 38(3):243–267.